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Perioperative antifibrotic therapy for patients with idiopathic pulmonary fibrosis undergoing lung cancer surgery: A systematic review and meta-analysis

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ABSTRACT

Background: Patients with idiopathic pulmonary fibrosis (IPF) undergoing lung cancer surgery face a 4.4–20 % risk of acute exacerbation (AE-IPF) with mortality exceeding 50 %. The potential role of perioperative antifibrotic therapy in reducing surgical complications in this high-risk population remains unclear.

Objectives: To evaluate whether perioperative antifibrotic therapy (pirfenidone/nintedanib) reduces complications, particularly acute exacerbations and mortality, in IPF patients undergoing lung cancer surgery through systematic review and meta-analysis.

Methods: Following PRISMA guidelines, we conducted a systematic review and meta-analysis of observational studies examining perioperative antifibrotic therapy in IPF patients undergoing lung cancer surgery. Four studies comprising 261 patients (124 treated, 137 controls) from Japan and Italy (2016–2024) were analyzed. Pooled risk ratios were calculated using Review Manager 5.4. The study protocol was registered with PROSPERO (ID: CRD42025649005).

Results: Perioperative antifibrotic therapy achieved a 69 % reduction in AE-IPF risk (RR 0.31, 95 % CI 0.13–0.70) and an 81 % reduction in 90-day mortality (RR 0.19, 95 % CI 0.07–0.52). Additional benefits included significantly shorter hospital stays (5 vs 7 days, $p = 0.029$) and reduced complications, including decreased prolonged air leak rates (3.4 % vs 26.9 %). Adverse events were minimal, consisting primarily of mild nausea and photosensitivity.

Conclusions: Perioperative antifibrotic therapy significantly reduces acute exacerbations and mortality in IPF patients undergoing lung cancer surgery. However, findings are limited by small observational studies concentrated in specific geographic regions. Randomized controlled trials are needed to confirm efficacy and establish standardized treatment protocols.

Abbreviation: ADC, Adenocarcinoma; AE-ILD, Acute Exacerbation of Interstitial Lung Disease; AE-IPF, Acute Exacerbation of Idiopathic Pulmonary Fibrosis; ALAT, Latin American Thoracic Association; ATS, American Thoracic Society; BMI, Body Mass Index; CCI, Charlson Comorbidity Index; CFIP, Chronic Fibrosing Interstitial Pneumonia; CI, Confidence Interval; COP, Cryptogenic Organizing Pneumonia; CRP, C-Reactive Protein; CTD-ILD, Connective Tissue Disease-associated Interstitial Lung Disease; DSS, Disease-Specific Survival; ERS, European Respiratory Society; FDA, Food and Drug Administration; FGF, Fibroblast Growth Factor; FVC, Forced Vital Capacity; GGO, Ground Glass Opacities; HP, Hypersensitivity Pneumonitis; HRCT, High-Resolution Computed Tomography; ILD, Interstitial Lung Disease; IPF, Idiopathic Pulmonary Fibrosis; IQR, Interquartile Range; IV, Inverse Variance; JACS, Japanese Association for Chest Surgeons; JRS, Japanese Respiratory Society; KL-6, Krebs von den Lungen-6; LDH, Lactate Dehydrogenase; NSCLC, Non-Small Cell Lung Cancer; NSIP, Non-Specific Interstitial Pneumonia; PAT, Perioperative Antifibrotic Therapy; PDGF, Platelet-Derived Growth Factor; P/F Ratio, Partial Pressure of Arterial Oxygen to Fraction of Inspired Oxygen Ratio; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; PROSPERO, International Prospective Register of Systematic Reviews; rhTM, Recombinant Human Thrombomodulin; RR, Risk Ratio; SCC, Squamous Cell Carcinoma; SE, Standard Error; SP-D, Surfactant Protein D; TGF- β , Transforming Growth Factor-Beta; TNF- α , Tumor Necrosis Factor-Alpha; UIP, Usual Interstitial Pneumonia; VEGF, Vascular Endothelial Growth Factor; WBC, White Blood Cell.

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Introduction

Idiopathic pulmonary fibrosis (IPF) is the most common form of interstitial lung disease, affecting 20–40 per 100,000 individuals worldwide and carrying a high mortality rate due to progressive lung dysfunction.^{1,2} Patients with IPF have a significantly elevated risk of developing lung cancer, with cumulative incidence rates reaching 20–30 % over the disease course, representing a 3.5 to 7.3-fold increase compared to age-matched controls.^{3,4} The co-existence of these conditions presents unique management challenges, as each disease influences the prognosis and treatment approach of the other.

Surgical resection remains the standard treatment for early-stage lung cancer, with five-year survival rates exceeding 70 % in stage I disease.⁵ However, IPF significantly complicates surgical management and increases perioperative risk.⁶ Postoperative acute exacerbation of IPF (AE-IPF) occurs in 4.4–20 % of cases,^{7,8} with a mortality rate exceeding 50 %.⁹ This high risk of life-threatening complications makes perioperative decision-making exceptionally complex for both clinicians and patients.³

Current treatment approaches for IPF focus on slowing disease progression through antifibrotic therapy, which has become the standard of care based on landmark clinical trials. Two FDA-approved antifibrotic agents (pirfenidone and nintedanib) have demonstrated significant efficacy: pirfenidone and nintedanib. The CAPACITY¹⁰ and ASCEND¹¹ trials showed that pirfenidone reduces the rate of forced vital capacity (FVC) decline and may reduce mortality risk, while the INPULSIS¹² trials demonstrated that nintedanib significantly reduces the annual rate of FVC decline compared to placebo. Both agents may also reduce the risk of acute exacerbations in chronic IPF management.¹³

These medications work through distinct but complementary mechanisms. Pirfenidone exerts its effects through multiple pathways, including inhibition of transforming growth factor-beta (TGF- β),¹⁴ platelet-derived growth factor (PDGF),¹⁴ and tumor necrosis factor-alpha (TNF- α),¹⁴ while also providing antioxidant¹⁵ and anti-inflammatory¹⁵ properties. Nintedanib, a tyrosine kinase inhibitor, targets multiple growth factor receptors including PDGF receptors,¹⁶ fibroblast growth factor (FGF) receptors,¹⁶ and vascular endothelial growth factor (VEGF) receptors,¹⁶ thereby interrupting key fibrotic pathways.

Both agents are recommended in current clinical guidelines for patients with IPF, with treatment typically initiated upon diagnosis in patients with mild-to-moderate disease (FVC $\geq$ 50 % predicted). These molecular targets become particularly relevant in perioperative settings, where surgical stress and tissue injury can trigger inflammatory cascades that may precipitate acute exacerbations. Beyond their established role in chronic disease management, emerging evidence suggests potential benefits in acute settings, including acute exacerbations,¹⁷ though this remains an area of active investigation. Emerging evidence suggests that perioperative antifibrotic therapy may offer protection by reducing the risk of AE-IPF risk and improve surgical outcomes, but findings have been inconsistent and limited by small sample sizes.^{18–22} To date, no comprehensive synthesis of this evidence has been undertaken to guide clinical practice.

This systematic review and meta-analysis aim to evaluate the impact of perioperative antifibrotic therapy in IPF patients undergoing lung cancer surgery, specifically assessing effects on AE-IPF rates, mortality, hospital length of stay, and overall survival.

Methods

Search strategy

A comprehensive systematic literature review was conducted by two independent investigators (N.S. and F.D.G.), spanning medical databases including MEDLINE, EMBASE, and the Cochrane libraries (both Central Register of Controlled Trials and Database of Systematic Reviews) from the inception of each database through January 2025. The

review focused on examining the perioperative use of antifibrotic agents, pirfenidone and nintedanib, in patients undergoing lung cancer surgery with concurrent IPF. The primary endpoint was the risk of AE-IPF, while secondary endpoints encompassed overall survival, length of hospital stay, and mortality rates. The search strategy encompassed the utilization of the following terms: "interstitial lung disease," "ILD," "Idiopathic pulmonary fibrosis," "IPF," "lung fibrosis," "fibrosing alveolitis," "Connective tissue disease-associated ILD," "CTD-ILD," "pulmonary pneumonitis," "pulmonary eosinophilia pneumonia," "Nonspecific interstitial pneumonia," "NSIP," "Cryptogenic organizing pneumonia," "COP," "Hypersensitivity pneumonitis," "Drug-induced ILD" combined with "antifibrotic," "nintedanib," "pirfenidone," in addition to terms signifying exacerbation such as "exacerbation," "disease exacerbation," "flare," and "flare up," as detailed in the supplementary data (see supplemental file 1). The reference lists of selected articles were manually reviewed for additional relevant studies based on the inclusion criteria, with no language restrictions applied. The systematic review followed PRISMA guidelines,²³ and the study protocol was registered in PROSPERO (ID: CRD42025649005).

Inclusion criteria

Eligible studies included clinical trials or observational studies (case-control or cohort designs) assessing the efficacy of antifibrotic medications when initiated perioperatively for lung cancer surgery in improving patient outcomes. Included studies also required a control group receiving standard treatment or no intervention, compared to patients receiving antifibrotics (various regimens and dosages), along with clear outcome reporting.

Exclusion criteria

Studies were excluded if they were non-comparative, lacked clear antifibrotic treatment details, did not define outcomes, or were case reports, case series or cross-sectional study. Additionally, studies that included patients who were already receiving antifibrotic therapy for IPF treatment at the time of cancer diagnosis or subsequent surgical resection were excluded. This exclusion criterion was implemented to specifically evaluate the effect of newly initiated perioperative antifibrotic therapy rather than the continuation of established treatment regimens, which could introduce confounding variables related to treatment duration and pre-surgical disease stabilization.

Data extraction and quality assessment

Data were extracted using a standardized form, capturing study title, primary author, publication year, country, antifibrotic intervention details (type, initiation timing, dose, regimen), outcome measures (survival, AE-IPF incidence, hospitalization duration, and mortality, sample size, demographics, and adjusted confounders with effect estimates and 95 % confidence intervals).

Two investigators (N.S. and F.D.G.) independently performed extraction and quality assessment using the Newcastle-Ottawa Scale,²⁴ which assesses selection (S), comparability (C), and exposure or outcome assessment (O). Studies scoring six or higher were considered good methodological quality. Detailed quality assessment results for each study are presented in Supplementary Table S1.

Statistical analysis

All statistical analyses were performed using Review Manager 5.4 (Cochrane Collaboration, London, UK). The pooled relative risk (RR) of AE-IPF who underwent lung cancer surgery who was initiated antifibrotic medications perioperatively, compared to those who did not receive antifibrotics, was calculated using a dichotomous approach based on raw data from the original studies. Statistical heterogeneity

was assessed using Cochran’s Q test, which is complemented with the I^2 statistics. This statistic quantifies the proportion of the total variation across studies, which is due to true heterogeneity rather than chance. A I^2 value of 0–25 % represents insignificant heterogeneity, >25 % but less than or equal to 50 % low heterogeneity, >50 % but less than or equal to 75 % moderate heterogeneity, and >75 % represents high heterogeneity.²⁵ Given the small number of studies ($n = 4$), formal publication bias assessment was not performed.

Results

Study selection and characteristics

Our systematic review process involved a thorough evaluation of 1785 potentially relevant articles identified through database searching and reference list screening. After removing 107 duplicates, 1678 unique studies underwent title and abstract screening. Through rigorous application of inclusion criteria, 1410 studies were excluded at the screening stage. Of the remaining 268 articles that underwent full-text review, 263 were ultimately excluded, with 196 eliminated due to inappropriate study designs (including narrative reviews, conference abstracts, case reports, and non-clinical studies) and 67 removed for inadequate outcome reporting (such as incomplete data, surrogate

endpoints, or insufficient follow-up periods). The selection comprised five observational studies but we identified potential overlap between the patient populations in Iwata et al. (2015)¹⁹ and Iwata et al. (2016),¹⁸ as the later publication incorporated patients from the earlier study with extended follow-up. Therefore, to prevent double-counting of outcomes, we included only data from Iwata et al. (2016) in our analysis. The final selection comprised four observational studies which collectively analyzed data from 261 patients (124 receiving perioperative antifibrotic therapy and 137 controls). Fig. 1 presents the PRISMA flow diagram detailing the study selection process. All included studies were observational designs, with four conducted in Japan^{18,21,22} and one in Italy.²⁰ Publication dates ranged from 2016 to 2024.

Pirfenidone was first approved for IPF in Japan in 2008 and later in Europe and the US, while nintedanib was approved in 2014. Most included studies utilized pirfenidone, reflecting its earlier adoption in clinical practice. Only one study reported the use of both pirfenidone and nintedanib, with most perioperative protocols established prior to the widespread availability of nintedanib.

Diagnosis of interstitial lung disease

As summarized in Table 1, methods for diagnosing IPF varied across

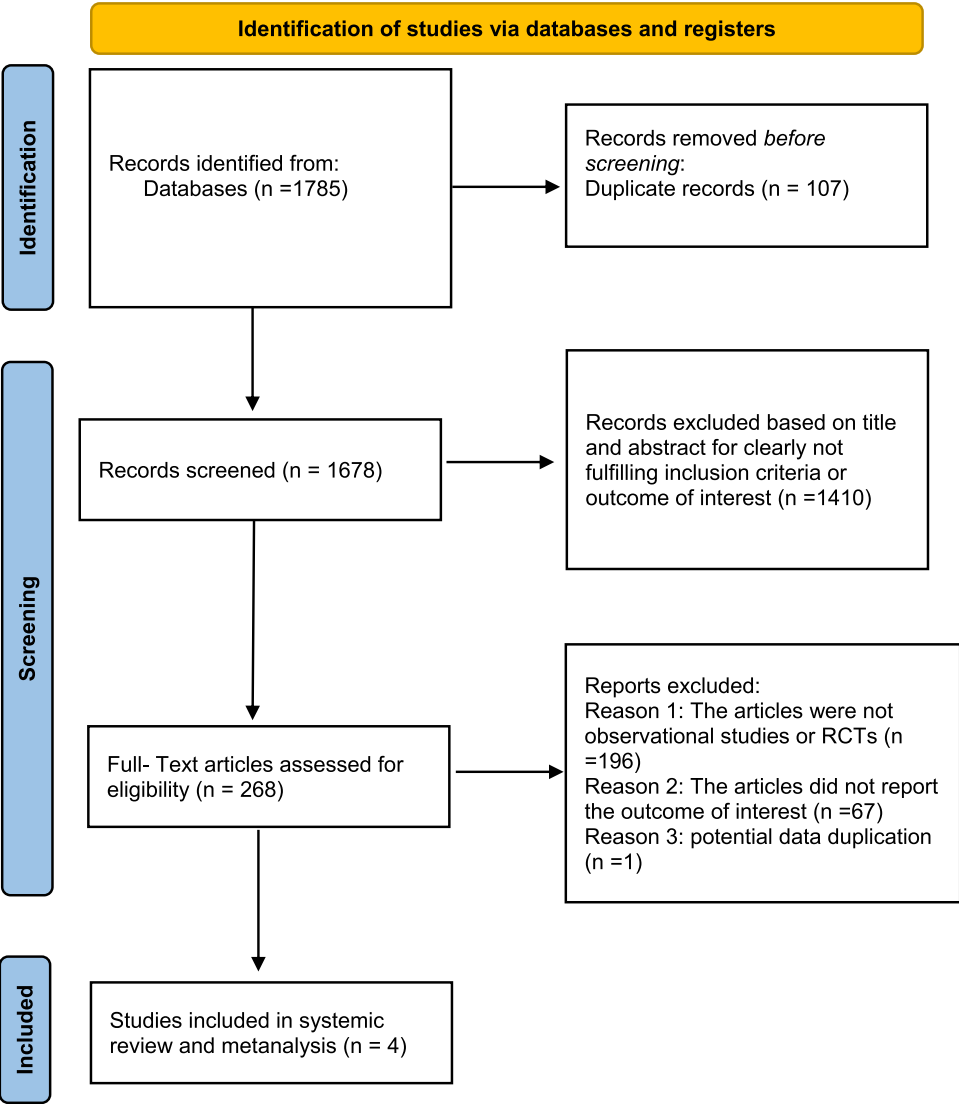


Fig. 1. Literature review process.

Table 1
Study characteristics.

Study	Country	Year	Study Design	Patient Selection	Controls group	IPF Diagnosis	Types of ILD	ILD Exacerbation Definition
Bongiolatti	Italy	2024	Retrospective cohort	Consecutive patients meeting inclusion criteria	used contemporaneous controls	Multidisciplinary discussion at ILD Referral Center	IPF	Acute, clinically significant respiratory deterioration with new, widespread alveolar abnormality, excluding alternative etiologies, developed <1 month after curative-intent lung surgery
Iwata	Japan	2016	Retrospective cohort	Consecutive patients with IPF undergoing surgery	historical controls from the pre-intervention period	-Radiology: HRCT confirmed IPF before pirfenidone treatment. -Exclusion: Other ILD causes were ruled out. -Pathology: UIP patterns in surgical cases confirmed IPF per 2011 ATS/ERS/JRS/ALAT criteria.	IPF	Subjective worsening of dyspnea, new bilateral radiologic opacities, no evidence of infection, exclusion of alternative causes (heart failure, pulmonary embolism)
Kanayama	Japan	2019	Retrospective cohort	Consecutive patients with IPF undergoing surgery	historical controls from the pre-intervention period	HRCT and pathological findings (ATS/ERS/JRS/LATA criteria)	IPF	Defined by ATS/ERS/JRS/LATA criteria: intensified dyspnea, increased interstitial shadow on imaging, >10 mmHg decrease in arterial oxygen tension, no evidence of infection, exclusion of alternative causes
Sekihara	Japan	2020	Retrospective cohort	Consecutive patients with IPF undergoing surgery	historical controls from the pre-intervention period	Based on HRCT according to specific criteria established by the Japanese Respiratory Society, which are consistent with the 2011 guidelines of the American Thoracic Society	IPF	Acute subjective worsening of dyspnea, new bilateral radiological opacities on HRCT, no evidence of infection, alternative causes excluded, decrease in arterial oxygen tension >10 mmHg, within 90 days post-surgery

Abbreviations: ATS, American Thoracic Society; ERS, European Respiratory Society; JRS, Japanese Respiratory Society; ALAT, Latin American Thoracic Association; LATA, Latin American Thoracic Association; ILD, Interstitial Lung Disease; IPF, Idiopathic Pulmonary Fibrosis; NSIP, Non-Specific Interstitial Pneumonia; HP, Hypersensitivity Pneumonitis; CTD-ILD, Connective Tissue Disease-associated Interstitial Lung Disease; HRCT, High-Resolution Computed Tomography; GGO, Ground Glass Opacities; CFIP, Chronic Fibrosing Interstitial Pneumonia; UIP, Usual Interstitial Pneumonia.

included studies. Bongiolatti et al.²⁰ employed a multidisciplinary discussion approach at a specialized referral center. Iwata et al.(2016)¹⁸ utilized multiple diagnostic criteria, including high-resolution computed tomography (HRCT) confirmation of IPF before initiating treatment with pirfenidone, pathological confirmation of usual interstitial pneumonia (UIP) patterns in surgical cases, and adherence to the 2011 American Thoracic Society (ATS), European Respiratory Society (ERS), Japanese Respiratory Society (JRS), and Latin American Thoracic Association (ALAT) guidelines. Sekihara et al.²² based their diagnosis on HRCT in accordance with JRS criteria, which aligned with the 2011 ATS guidelines. Kanayama et al.²¹ relied on HRCT and pathological findings for diagnosis.

Patient characteristics

Table 2 presents the demographic and clinical characteristics of patients included in the studies. Sample sizes varied considerably across studies, with the largest study (Kanayama et al.²¹) including 100 patients (28 in the antifibrotic group, 72 in the control group) and the smallest (Iwata et al.¹⁸) including 50 patients (31 receiving antifibrotics, 19 controls).

Patients were predominantly male across all studies, consistent with the epidemiology of both IPF and lung cancer. Mean/median ages ranged from 68 to 75.5 years. Baseline pulmonary function testing revealed similar lung function between treatment groups, though control groups tended to have slightly better FVC and diffusing capacity of the lung for carbon monoxide. Where reported, the predominant cancer types were squamous cell carcinoma and adenocarcinoma. Surgical approaches included lobectomy, segmentectomy, and wedge resection, though the distribution varied across studies.

Lung cancer histological characteristics

Lung cancer histological subtypes were incompletely reported across the included studies. Only Bongiolatti et al.²⁰ provided detailed histological classification, reporting squamous cell carcinoma in 50 % of the perioperative antifibrotic therapy (PAT) group versus 44.8 % in controls, and adenocarcinoma in 34.5 % of the PAT group versus 38.5 % in controls. The remaining three studies (Iwata et al.,¹⁸ Kanayama et al.,²¹ and Sekihara et al.²²) did not specify lung cancer histological subtypes, representing 75 % of the total study population (*n* = 206 of 261 patients). This limitation in histological characterization may affect the generalizability of findings across different lung cancer subtypes, as various histologies may exhibit different inflammatory profiles and responses to perioperative antifibrotic therapy.

Antifibrotic protocol variations and dosing strategies

Table 3 details the antifibrotic protocols used in each study, revealing significant variation in dosing strategies that reflects the lack of standardized protocols for perioperative use. The most common approach was a graduated dosing strategy for pirfenidone, employed by two studies (Iwata et al.¹⁸ and Kanayama et al.²¹). This protocol began with 600 mg daily for the first week, escalated to 1200 mg daily (divided into two doses) for the second week, and reached the target dose of 1800 mg daily (600 mg three times daily) by the third week when tolerated. This gradual escalation mimics the approach used in chronic IPF management and helps minimize gastrointestinal side effects. In contrast, Bongiolatti et al.²⁰ used both pirfenidone and nintedanib, with pirfenidone administered at a higher maintenance dose of 2403 mg daily (801 mg three times daily), which exceeds the standard chronic dosing, while

Table 2
Patient characteristics.

Study	Group	Sample Size	Sex (Female %)	Age (years)	Lung Cancer Type	Type of lung resection	Smoking (%)	CCI	PFT
Bongiolatti	PAT	29	27.6	72 (66–77)*	SCC: 50 %, ADC: 34.5 %	Not specified	Not reported	5 (4–6)*	FEV1 % 88 FVC % 89 DLCO % 58
	Non-PAT	26	23.1	75.5 (66–78)*	SCC: 44.8 %, ADC: 38.5 %	Not specified	Not reported	6 (4–6.5)*	FEV1 % 96.5 FVC % 98 DLCO % 65.5
Iwata	PAT	31	9.7	68.0 (36–82)†	Not specified type	Anatomical resection: 23 Wedge resection: 8	90.3	Not reported	FVC % 94.5 DLCO % 74.3
	Non-PAT	19	5.3	68.7 (43–83)†	Not specified type	Anatomical resection: 18 Wedge resection: 1	94.7	Not reported	FVC % 102.7 DLCO % 80.8
Kanayama	PAT	28	7.1	73.7 (68.3–79.1)†	Not specified type	Anatomical resection: 24 Wedge resection: 4	Not reported	Not reported	FVC % 92.7 FEV1 % 74.1
	Non-PAT	72	13.9	72.6 (65.6–79.6)†	Not specified type	Anatomical resection: 59 Wedge resection: 13	Not reported	Not reported	FVC % 96.6 FEV1 % 72.4
Sekihara	PAT	36	11	72 (51–86)†	Not specified type	Wedge resection: 3 Segmentectomy: 0 Lobectomy: 31 Pneumonectomy:	Current: 42 %, Ex: 58 %	Not reported	VC % 94 FEV1 % 79
	Non-PAT	20	10	72 (60–81)†	Not specified type	Wedge resection: 2 Segmentectomy: 1 patient Lobectomy: 12 Pneumonectomy: 1	Current: 30 %, Ex: 70 %	Not reported	VC % 97 FEV1 % 79

*Median (IQR) †Mean (range).
Abbreviations: PAT, Perioperative Antifibrotic Treatment; SCC, Squamous Cell Carcinoma; ADC, Adenocarcinoma; IQR, Interquartile Range; CCI, Charlson Comorbidity Index; Ex, Ex-smoker; PFT, Pulmonary Function Test; VC, Vital Capacity; FVC, Forced Vital Capacity; FEV1, Forced Expiratory Volume in 1 s; DLCO, Diffusing Capacity of the Lung for Carbon Monoxide.

Table 3
Treatment characteristics.

Study	Medication	Dosage	Timing	Compared Groups	Confounder Adjustments
Bongiolatti	Pirfenidone (primary), Nintedanib	Pirfenidone: 801 mg TID (2403 mg/day), Nintedanib: not specified	Pirfenidone: 2–4 weeks pre-surgery, Nintedanib: paused 2–5 days pre-surgery	no-treatment group	DLCO %, GAP index, AFT use, Age, sex, BMI, FEV1 %, FVC %, ECOG status
Iwata	Pirfenidone	Starting: 600 mg/day, increased to 1200 mg/day, then 1800 mg/day if possible	4 weeks before to 4 weeks after operation	non-PAT group	JACS risk score (including AE history, resection type, UIP pattern, steroid usage, sex, KL-6 levels, vital capacity)
Kanayama	Pirfenidone	Starting: 600 mg/day, increased to 1200 mg/day	2–4 weeks before surgery	non-PAT group	Adjusted for JACS risk scores by subdividing into risk groups
Sekihara	Pirfenidone	Not specified	Perioperative period	non-PAT group	specific adjustments not detailed

Abbreviations: BMI, Body Mass Index; CCI, Charlson Comorbidity Index; AE-ILD, Acute Exacerbation of Interstitial Lung Disease; P/F Ratio, Partial Pressure of Arterial Oxygen to Fraction of Inspired Oxygen Ratio; WBC, White Blood Cell; CRP, C-Reactive Protein; LDH, Lactate Dehydrogenase; KL-6, Krebs von den Lungen-6; SP-D, Surfactant Protein D; rhTM, Recombinant Human Thrombomodulin; AE-IPF, Acute Exacerbation of Idiopathic Pulmonary Fibrosis; JACS, Japanese Association for Chest Surgeons Score; TID, Three Times Daily; DLCO %, Diffusing Capacity of the Lungs for Carbon Monoxide (percentage of predicted); GAP, Gender-Age-Physiology; AFT, Antifibrotic Therapy; FEV1 %, Forced Expiratory Volume in 1 s (percentage of predicted); FVC %, Forced Vital Capacity (percentage of predicted); ECOG, Eastern Cooperative Oncology Group; PAT, Perioperative Antifibrotic Treatment.

the nintedanib dosage was not specified. Sekihara et al.²² also used pirfenidone but did not specify the dosage. All studies compared their treatment groups against non-treatment control groups. The timing of initiation varied, with most studies starting treatment 2–4 weeks pre-operatively and continuing for 4–8 weeks postoperatively. This perioperative window was designed to establish therapeutic levels before surgical stress while maintaining protection during the high-risk postoperative period. **Primary Outcome**

Acute exacerbation of idiopathic pulmonary fibrosis

All four studies reported the incidence of AE-IPF after lung cancer surgery, though the timing of assessment varied (30-day, 90-day, or unspecified postoperative period). As shown in Table 4, all studies consistently reported lower AE-IPF rates in patients receiving PAT compared to controls.
Bongiolatti et al.²⁰ demonstrated a significant reduction in AE-IPF

Table 4

Treatment outcomes.

Study	AE Incidence	In hospital mortality	30-day Mortality	90-day Mortality	Length of Stay days	Survival	Other Significant Outcomes	Quality assessment
Bongiolatti	3.4%vs. 23.1% (p=0.044)	Not reported	3.4%vs. 7.7% (p=0.601)	6.9%vs. 15.4% (p=0.412)	5 vs. 7 days (p=0.029)	3-year: 55.2% vs. 42.3%	- Lower pulmonary complications (31%vs 50%, p=0.145) - Reduced prolonged air leak (3.4%vs 26.9%, p=0.021) - - Few adverse events (mild nausea and photosensitivity) - Lower estimated blood loss in PAT group - Trend toward better progression-free survival (not significant)	S4, C2, O3
Iwata	0%vs. 10.5% (30D), 3.2% vs. 21.1% (90D)	Not reported	PAT group: 0% (0 out of 31 patients) Non-PAT group: 10.5% (2 out of 19 patients)	PAT group: 3.2% (1 out of 31 patients) Non-PAT group: 21.1% (4 out of 19 patients)	Not reported	Not significant (p=0.0676)	Preoperative adverse events related to PAT: 14.2% (4/28) Postoperative complications (excluding AE):10.7% vs. 13.7%	S4, C2, O3
Kanayama	3.6%vs. 13.9% (30 days) 3.6%vs. 16.7% (90 days)	Not reported	Not reported	Not reported	Not reported	1-year DSS: 83.0% vs. 83.7%; 3-year DSS: 66.4% vs. 47.8%	No severe adverse events from pirfenidone were observed -Longer median interval to AE-IPF in PAT group (23 vs 9 days, p=0.03)	S4, C2, O3
Sekihara	8%vs. 20% (p=0.23)	Not reported	0%vs. 15% (p=0.04)	3%vs 20% (p=0.05)	14 vs. 15 days (p=0.01)	3-year: 37% vs. 47% (p=0.82)		S4, C0, O3

Abbreviations: AE, Acute Exacerbation; DSS, Disease-Specific Survival; PAT, Perioperative Antifibrotic Treatment; KL-6, Krebs von den Lungen-6; IPF, Idiopathic Pulmonary Fibrosis; AEs, Adverse Events; S, Selection; C, Comparability; O, Outcome; AE-IPF, Acute Exacerbation of Idiopathic Pulmonary Fibrosis.

(3.4 % in the PAT group versus 23.1 % in controls, $p = 0.044$). Similarly, Iwata et al.¹⁸ reported substantially lower rates of AE-IPF at both 30 days (0 % versus 10.5 %) and 90 days (3.2 % versus 21.1 %). Kanayama et al.²¹ observed AE-IPF in 3.6 % of PAT patients compared to 13.9 % of controls at 30 days, with the same rates at 90 days (3.6 % versus 16.7 %). While Sekihara et al.²² found a similar trend (8 % versus 20 %), this difference did not reach statistical significance ($p = 0.23$).

Secondary outcomes

Mortality and survival

Thirty-day mortality rates varied across studies, with Bongiolatti et al.²⁰ showing 3.4 % versus 7.7 % ($p = 0.601$) and Sekihara et al.²² demonstrating 0 % versus 15 % ($p = 0.04$) in PAT and control groups, respectively. Ninety-day mortality rates showed similar patterns of reduction.

Length of hospital stay and complications

The hospital length of stay was reported by Bongiolatti et al.,²⁰ who found a significant reduction in PAT group (median 5 days versus 7 days, $p = 0.029$). Two studies^{20,21} observed decreased postoperative complications in patients receiving antifibrotics. Bongiolatti et al.²⁰ reported lower pulmonary complication rates (31 % vs. 50 %, $p = 0.145$) and significantly reduced prolonged air leak incidence (3.4 % vs. 26.9 %, $p = 0.021$), while Kanayama et al.²¹ noted fewer overall postoperative complications in the PAT group (10.7 % versus 13.7 %, though statistical significance was not reported). Interestingly, Iwata et al.¹⁸ reported reduced intraoperative blood loss in the PAT group.

Long-term survival

Several studies reported long-term survival but with varying timeframes and outcomes. Bongiolatti et al.²⁰ reported three-year survival rates of 55.2 % versus 42.3 % in antifibrotic and control groups, respectively, though statistical significance was not reported. Sekihara et al.²² found no significant difference in survival (37 % versus 47 % at an unspecified time point, $p = 0.82$).

Safety and adverse effects

The studies consistently reported mild side effects, primarily nausea and photosensitivity. Kanayama et al.²¹ reported preoperative adverse events related to PAT in 14.2 % of patients, though these did not necessitate surgical delays or treatment discontinuation. No severe adverse events were reported in any study.

Meta-Analysis results

Our meta-analysis evaluated two key outcomes across the four included studies (comprising 261 patients: 124 receiving PAT and 137 controls). As detailed in Table 5, we analyzed both the incidence of AE-IPF and 90-day mortality.

For AE-IPF risk (using 90-day data when both 30- and 90-day results were available), we found a pooled risk ratio (RR) of 0.31 (95 % CI, 0.13 - 0.70) for PAT compared to controls. Statistical heterogeneity was insignificant ($I^2 = 0$ %), suggesting consistency in the protective effect across studies despite methodological differences. Fig. 2 presents the forest plot of the included studies.

Similarly, our meta-analysis of 90-day mortality yielded a pooled RR of 0.19 (95 % CI, 0.07–0.52) for patients receiving PAT compared to controls. Statistical heterogeneity was again insignificant ($I^2 = 0$ %), further supporting the consistency of findings across studies. Fig. 3 demonstrates the forest plot for mortality outcomes.

Subgroup and sensitivity analyses

Due to the limited number of included studies and homogeneity in reported outcomes (as evidenced by $I^2 = 0$ %), formal subgroup analyses based on antifibrotic type, timing, or surgical approach were not feasible. Additionally, direct comparison between pirfenidone and nintedanib was not possible given the sparse nintedanib data, which was used in only one study by Bongiolatti et al., where both agents were utilized in the treatment group. With most evidence (3 of 4 studies) derived from pirfenidone use, drug-specific subgroup analysis could not be conducted.

Table 5
Acute exacerbation of interstitial lung disease events and mortality in antifibrotic vs. non-antifibrotic treatment groups.

Study	Antifibrotic group		Non antifibrotic group		Antifibrotic group		Non antifibrotic group	
	Events (AE-IPF)	Total patients	Events (AE-IPF)	Total patients	Events (Death)	Total patients	Events (Death)	Total patients
Bongiolatti	1	29	6	26	0	29	3	26
Iwata	1	31	4	19	3	31	8	19
Kanayama	2	28	12	72	Not report	Not report	Not report	Not report
Sekihara	3	36	4	20	1	36	4	20
Total	7	124	26	137	4	96	15	82

Abbreviations: AE-IPF, Acute Exacerbation of Interstitial Pulmonary Fibrosis.

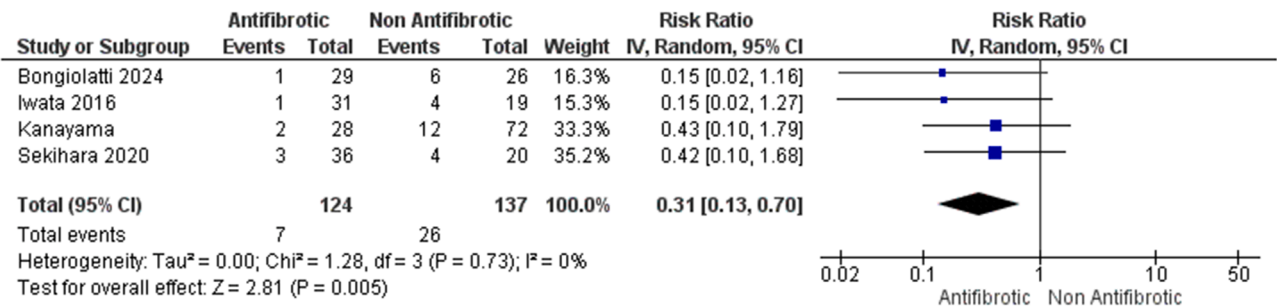


Fig. 2. Forest plot of the included studies comparing the risk of acute exacerbation in interstitial lung disease in patients who received antifibrotic and those without antifibrotic medication; square data markers represent risk ratios (RR). Horizontal lines, the 95 % confidence interval (CI) with marker size reflecting the statistical weight of the study using random-effects meta-analysis. A diamond data marker represents the overall RR and 95 % CI for the outcome of interest. IV, inverse variance; SE, standard error.

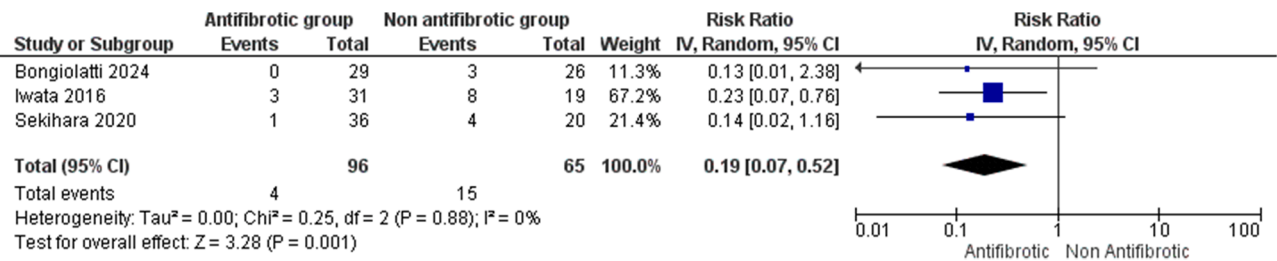


Fig. 3. Forest plot of the included studies comparing mortality risk in patients who received antifibrotic and those without antifibrotic medication; square data markers represent risk ratios (RR). Horizontal lines, the 95 % confidence interval (CI) with marker size reflecting the statistical weight of the study using random-effects meta-analysis. A diamond data marker represents the overall RR and 95 % CI for the outcome of interest. IV, inverse variance; SE, standard error.

Discussion

Main findings and clinical significance

Our systematic review and meta-analysis provide evidence supporting the beneficial role of PAT in IPF patients with coexisting lung cancer undergoing lung cancer surgery. The findings demonstrate significant reductions in AE-IPF and mortality rates, with a consistent safety profile across studies.

The meta-analysis revealed that PAT reduced the risk of AE-IPF by 69 % (RR 0.31, 95 % CI 0.13–0.70) compared to standard care without antifibrotics. This substantial reduction in AE-IPF is particularly noteworthy given that these events traditionally carry mortality rates exceeding 50 %.⁹ Consistent findings across several studies with similar populations and intervention types suggest potential generalizability, with further studies needed to confirm these findings in different settings. Similarly, the significant reduction in 90-day mortality (RR 0.19, 95 % CI 0.07–0.52) suggests that PAT may have a mortality benefit in this high-risk patient population. This improvement in survival, coupled with the reduced length of hospital stay and decreased complication rates reported in several studies, indicates that the benefits of PAT may extend beyond preventing acute exacerbations to broader improvements in perioperative outcomes.

Mechanisms of action and therapeutic implications

The mechanism of action of the studied antifibrotic agents helps explain their potential effectiveness in the perioperative setting. Pirfenidone exhibits multiple effects, including antifibrotic,¹⁵ anti-inflammatory,¹⁵ and antioxidant properties.¹⁵ It works by inhibiting TGF- β ,¹⁴ PDGF,¹⁴ and TNF- α ,¹⁴ all of which play crucial roles in the development and progression of pulmonary fibrosis.^{14,15} Nintedanib, a tyrosine kinase inhibitor,¹⁶ targets multiple growth factor receptors including PDGF receptors,^{16,26} FGF receptors,^{16,26} and VEGF receptors.^{16,26} These molecular targets are particularly relevant in the context of surgical stress,⁸ where tissue injury and subsequent healing responses can trigger inflammatory cascades that may precipitate AE-IPF.²⁷ By modulating these pathways, antifibrotic agents may dampen the inflammatory response to surgical injury and prevent the cascade that leads to AE-IPF. The complex interplay between these therapeutic targets and the pathophysiological mechanisms underlying perioperative AE-IPF is illustrated in Fig. 4, which demonstrates how pirfenidone and nintedanib target key inflammatory and fibrotic cascades triggered by surgical stress.

Several explanations have been proposed for why these antifibrotic agents might be effective even after just a few weeks of perioperative use. The rapid onset of action may be attributed to their ability to

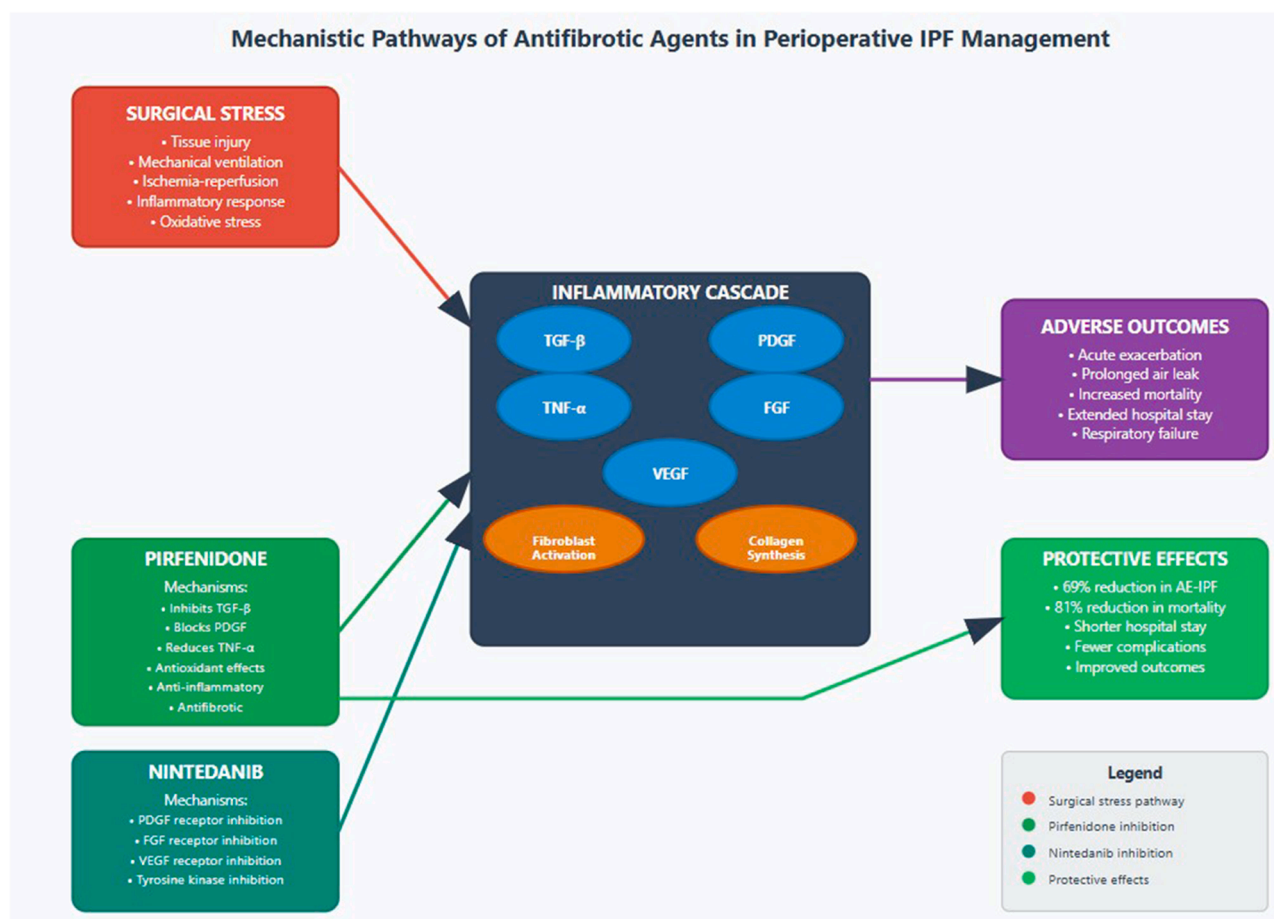


Fig. 4. Schematic showing how pirfenidone and nintedanib interrupt the surgical stress-induced inflammatory cascade involving transforming growth factor-beta (TGF- β), platelet-derived growth factor (PDGF), tumor necrosis factor-alpha (TNF- α), fibroblast growth factor (FGF), and vascular endothelial growth factor (VEGF) to prevent fibroblast activation and reduce adverse perioperative outcomes in patients with idiopathic pulmonary fibrosis.

quickly inhibit key inflammatory mediators activated during surgical stress. Studies have shown that pirfenidone treatment results in significantly decreased serum KL-6 levels,¹⁹ a marker associated with lung fibrosis, before and after surgery compared to control groups. This suggests that even short-term use can impact disease activity and potentially prevent post-surgical complications. Historical evidence also supports the use of antifibrotics during active acute exacerbations, with some studies indicating that initiating treatment during these critical periods can help stabilize patients.^{28,29} The strategic timing of administration, starting four weeks before surgery and continuing for four weeks afterward, appears to create a protective window during which the drug can exert its effects before and during the period of maximum surgical stress. Additionally, histopathological findings have revealed lower scores for pathological features associated with IPF in treated groups,¹⁹ suggesting that brief PAT might attenuate tissue damage normally seen during the surgical period.

The observed reduction in complications such as prolonged air leaks suggests that these agents might also modulate the wound healing response in ways that extend beyond their primary antifibrotic effects. This could be particularly relevant in the context of lung surgery, where tissue healing at resection margins and staple lines is critical for preventing air leaks and other complications.

Historical controls and surgical approach

Most included studies (3 of 4) utilized historical controls rather than contemporaneous control groups, which may introduce selection bias. This is particularly concerning given the wide range in postoperative

AE-IPF rates (10 % to 23 %) observed in control groups across studies. During the time covered by these studies, improvements in perioperative care, anesthetic techniques, and surgical approaches may have occurred independently of antifibrotic use, potentially confounding the observed effects. We attempted to analyze outcomes based on surgical approach (lobectomy versus segmentectomy versus wedge resection), but the studies did not consistently report outcomes stratified by procedure type, preventing such subgroup analysis. This represents an important area for future research, as the extent of surgical resection may influence both the risk of acute exacerbation and the potential protective effect of antifibrotic therapy.

Long-term survival analysis

While several studies reported long-term survival outcomes, these results should be interpreted with extreme caution. Cancer stage, histological type, and adjuvant treatments likely exert stronger influences on long-term survival than perioperative antifibrotic use. Furthermore, it remains unclear how many patients in the control groups subsequently received antifibrotic therapy as part of their standard IPF care, which would substantially confound any long-term comparisons. For these reasons, we consider the short-term outcomes (90-day AE-IPF rates and mortality) to be more reliable indicators of the specific effects of perioperative antifibrotic therapy.

Clinical implications and implementation

These findings have important clinical implications, as the potential

reduction in AE-IPF and mortality supports the consideration of PAT in IPF patients undergoing lung cancer surgery. However, all included studies were observational, and the number of study subjects was relatively modest and included mostly Japanese patients. Further research is needed to assess the exact role of antifibrotic therapy for patients with IPF and other fibrotic interstitial lung diseases undergoing lung cancer surgery and refine treatment protocols.

Strength and limitations

Strengths of our systematic review include a comprehensive search strategy across three databases (MEDLINE, EMBASE, and Cochrane libraries), combined with a rigorous formal review process using predetermined eligibility criteria. Our strict inclusion and exclusion criteria effectively ensured greater homogeneity among the study populations included.

Several limitations should be acknowledged. First, the analysis was limited by the small number of included studies, with only four observational studies meeting the criteria. Second, all included studies were published in English, introducing potential selection bias. Third, three of the four included studies were conducted in Japan, with only one from Italy, raising concerns about the generalizability of our findings to populations with different genetic backgrounds, environmental exposures, and healthcare systems. Japanese patients may have distinct IPF progression patterns, surgical techniques, and medication responses that may not directly apply to other populations. Fourth, all included studies focused specifically on IPF, which significantly limits the extrapolation of our findings to patients with other ILD subtypes. Fifth, while both pirfenidone and nintedanib were studied, most of the evidence reflected pirfenidone use, with variability in dosing regimens. Sixth, the optimal timing and duration of perioperative antifibrotic therapy remain uncertain due to inconsistent protocols. Seventh, the observational design inherently introduces potential bias; the limited number of studies makes publication bias difficult to assess. Eighth, the study population consistently had normal or near-normal mean percent-predicted FVC values, ranging from 82 % to 95 %. Since AE-IPF occurs less commonly in patients with preserved lung function, our findings may not be generalizable to individuals with more advanced disease who might be at higher risk for postoperative complications. Ninth, an important limitation of the current evidence base is the exclusion of patients already receiving antifibrotic therapy at the time of cancer diagnosis. Our inclusion criteria specifically excluded studies that included patients on established antifibrotic treatment, focusing instead on newly initiated perioperative therapy. This approach was chosen to isolate the specific effects of perioperative initiation rather than continuation of existing therapy. However, this creates a significant knowledge gap regarding outcomes in patients already on antifibrotic therapy, who represent a substantial portion of IPF patients with lung cancer. In clinical practice, many IPF patients are already receiving pirfenidone or nintedanib when lung cancer is diagnosed, and decisions about continuing, modifying, or temporarily discontinuing these agents perioperatively remain unclear. Future studies should specifically address outcomes in patients continuing established antifibrotic therapy versus those with perioperative modifications, as this represents a more clinically relevant scenario for many patients. Additionally, differences in IPF diagnostic criteria and definitions of acute exacerbation across studies may impact the interpretation of pooled results.

Nonetheless, given strict inclusion and exclusion criteria with a focused assessment of only perioperative antifibrotic use, and no other type of ILD reported, our analysis resulted in minimal heterogeneity.

Future research directions

Future research should explore several critical areas: First, determining optimal dosing strategies and timing of perioperative antifibrotic therapy through randomized controlled trials. Second, investigating the

potential benefits in non-IPF forms of ILD. Third, conducting head-to-head comparisons of pirfenidone versus nintedanib efficacy in the perioperative setting. Fourth, developing hybrid imaging approaches that combine high-resolution computed tomography with functional imaging to better predict which patients would benefit most from perioperative antifibrotic therapy. Fifth, integrating machine learning algorithms to analyze perioperative risk factors and optimize patient selection for antifibrotic therapy. Sixth, understanding the mechanistic basis for reduced complications through biomarker studies and histopathological analysis. Finally, evaluating long-term outcomes beyond the immediate perioperative period, including quality of life measures and disease progression rates. Randomized controlled trials provide the highest level of evidence, and we note that such a study assessing the role of perioperative pirfenidone therapy in patients with IPF and lung cancer is underway.³⁰

Conclusion

Our systematic review and meta-analysis provide evidence that perioperative antifibrotic therapy may reduce acute exacerbations and mortality in patients with IPF undergoing lung cancer surgery. Additional studies are needed to confirm this beneficial effect and define optimal treatment protocols for IPF and other fibrotic ILDs.

CRediT authorship contribution statement

Narat Srivali: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Federica De Giacomi:** Data curation. **Teng Moua:** Writing – review & editing, Supervision. **Jay H. Ryu:** Writing – review & editing, Supervision.

Declaration of competing interest

All authors do not have any financial or non-financial potential conflicts of interest.

Disclosure statement

The authors declare no conflicts of interest related to this work.

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Supplementary materials

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